Laboratory 2 Report Template:

Signals and Systems
Signal Transformations in Hardware
Time Shift, Amplitude Scaling, and Time Reversal

Graded Submission Requirement

Deliverable (1/11): This is a graded assignment. Include your name and student ID in every required screenshot and in as many additional screenshots as possible. Screenshots without clear student identification may not receive full credit.

Checkpoint (1/7): Before submission, verify that each screenshot is legible and contains visible labels/measurements where required.

Student Information

Name 1	______________________	**Name 2**	______________________
NetID 1	______________________	**NetID 2**	______________________
Section	______________________	**Date**	______________________
TA	______________________	**Random Seed**	______________________

Note on Prompts

- **Pre-lab Deliverable (N):** Pre-lab work to complete before lab.
- **Checkpoint (N):** Items to show your TA in lab.
- **Discussion (N):** Analysis and interpretation in your report.
- **Deliverable (N):** Required report artifacts and evidence.

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1 Assigned Parameters

Table 1. TA-Assigned Base Signal Parameters

Parameter	Description	Assigned Value
A	Base signal amplitude (V)	__________
f_0	Fundamental frequency (Hz)	__________
ϕ	Initial phase (rad)	__________
α	Exponential decay rate (s^{-1}), if applicable	__________
Signal Type	Sinusoidal / Exponential / Damped Sinusoidal	__________
t_0	Time shift duration (ms)	__________
B	Amplitude scale factor	__________

2 Pre-Lab Results

Pre-lab Deliverable (1/1): Attach your complete pre-lab pages or include concise summaries below.

2.1 Transformation Derivations

- $y_1(t) = x(t - t_0)$: ______________________________________
- $y_2(t) = B\,x(t)$: ______________________________________
- $y_3(t) = x(-t)$: ______________________________________
- $y_4(t) = B\,x(-t - t_0)$: ________________________________

2.2 Discrete-Time Mapping

- Shift depth: $k = \lfloor t_0/T_s \rfloor =$ ________ samples
- MATLAB mapping for y_1, y_2, y_3, y_4: __________________________________

2.3 Sketch Summary

Insert pre-lab sketch image (x, y1, y2, y3, y4) or attach scanned page

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3 Phase 1: Base Signal Acquisition

Checkpoint (2/7): Verify base signal generation and acquisition in both scope and MATLAB.

3.1 Step 1.1 Base Signal on Oscilloscope

Deliverable (2/11): Submit scope screenshot with name/ID, time/div, volts/div, measured frequency, measured amplitude.

Table 2. Measured Base Signal Parameters

Measured frequency (Hz)	____________________
Measured peak-to-peak voltage (Vpp)	____________________
Time/div	____________________
Volts/div	____________________

Step 1.1 Evidence: Oscilloscope screenshot (base signal)

3.2 Step 1.2 MATLAB Base Signal Acquisition

Deliverable (3/11): Submit MATLAB plot of x_hw with labeled axes and title.

Step 1.2 Evidence: MATLAB base-signal plot screenshot

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Discussion (1/7): Compare scope and MATLAB base signal. Report amplitude error and likely cause.

$$\varepsilon_A(\%) = \frac{|A_{scope} - A_{MATLAB}|}{A_{scope}} \times 100 = \underline{\hspace{3cm}}\%$$

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4 Phase 2: Hardware Transformations

4.1 Step 2.1 Amplitude Scaling

Checkpoint (3/7): Implement gain control with potentiometer and verify B ratio.**Deliverable (4/11):** Submit dual-channel scope screenshot (scaled vs original) with annotated ratio.

Table 3. Amplitude Scaling Measurements

A_{Ch1} (scaled)	
A_{Ch2} (original)	
$B_{meas} = A_{Ch1}/A_{Ch2}$	
Assigned B	
Percent error in B	

Step 2.1 Evidence: Scope screenshot for amplitude scaling

Discussion (2/7): Why is potentiometer reading performed outside ISR? Why must shared gain variable be volatile?

4.2 Step 2.2 Time Shift

Checkpoint (4/7): Implement circular buffer delay with depth k and verify time shift.**Deliverable (5/11):** Submit dual-channel scope screenshot with cursor-measured delay.

Table 4. Time Shift Measurements

Predicted t_0 (ms)	
Measured delay (ms)	
Absolute error (ms)	
Percent error	

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Step 2.2 Evidence: Scope screenshot for time shift

Discussion (3/7): Explain time delay vs phase shift equivalence for sinusoid and why it does not generalize to all signal classes.

4.3 Step 2.3 Time Reversal

Checkpoint (5/7): Implement reversed LUT indexing and verify waveform reversal.**Deliverable (6/11):** Submit dual-channel scope screenshot with key distinguishing feature annotated.

Step 2.3 Evidence: Scope screenshot for time reversal

Discussion (4/7): Classify x(t) as even/odd/neither and explain expected reversal behavior.

4.4 Step 2.4 Combined Transformation

Checkpoint (6/7): Implement and verify $y_4(t) = B\,x(-t - t_0)$.**Deliverable (7/11):** Submit dual-channel scope screenshot with ratio and time-offset annotations.

Step 2.4 Evidence: Scope screenshot for combined transformation

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5 Phase 3: MATLAB Verification

5.1 Step 3.1 Acquire All Signals

Checkpoint (7/7): Acquire x, y1, y2, y3, y4 into MATLAB.**Deliverable (8/11):** Submit five MATLAB plots with proper labeling.

Step 3.1 Evidence: MATLAB screenshots for x, y1, y2, y3, y4

5.2 Step 3.2 Predicted vs Measured Overlays

Deliverable (9/11): Submit four overlays: base, hardware, predicted for each transformation.

Step 3.2 Evidence: MATLAB overlay screenshots (y1-y4)

5.3 Step 3.3 NRMSE Quantification

Deliverable (10/11): Report NRMSE for y1, y2, y3, y4 using the specified formula.

Table 5. NRMSE Summary

Time Shift (y_1) NRMSE (%)	____________________
Amplitude Scaling (y_2) NRMSE (%)	____________________
Time Reversal (y_3) NRMSE (%)	____________________
Combined (y_4) NRMSE (%)	____________________

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Step 3.3 Evidence: MATLAB output/table screenshot for NRMSE

Discussion (5/7): Which transformation has largest NRMSE and why? Separate systematic vs random error sources.

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6 Data Analysis

6.1 Transformation Summary

Table 6. Transformation Summary Table

Parameter	Theoretical	Measured	% Error
Base amplitude A (V)			
Base frequency f_0 (Hz)			
Scaled amplitude $B \cdot A$ (V)			
Gain ratio B_{meas}			
Time shift t_0 (ms)			
Phase shift ϕ_0 (rad)			
Combined amplitude $B \cdot A$ (V)			
Combined shift t_0 (ms)			

Discussion (6/7): Identify the largest discrepancy source for each transformation and justify with measured evidence.

6.2 Symmetry and Boundary Effects

Discussion (7/7): Classify signal symmetry (even/odd/neither) and discuss `circshift` wrap-around artifacts with mitigation strategy.

6.3 Error Source Ranking

Deliverable (11/11): Rank top three contributors to error using NRMSE and physical justification (PWM quantization, RC phase distortion, ADC quantization, UART jitter, potentiometer noise, ISR jitter, etc.).

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Final Submission Checklist

- ☐ Pre-lab derivations and sketches included.
- ☐ Step 1.1 scope screenshot with name/ID.
- ☐ Step 1.2 MATLAB base signal plot.
- ☐ Step 2.1-2.4 scope screenshots with annotations.
- ☐ Step 3.1 five MATLAB signal plots.
- ☐ Step 3.2 four overlay plots.
- ☐ Step 3.3 NRMSE table/screenshot.
- ☐ Completed analysis tables and discussion responses.

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